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Web Technology

Quality Laboratory Manual

Experiment No. 01

Program Based on newly introduced elements of HTML5



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Experiment No. 01

Title of Experiment: Program Based on newly introduced elements of HTML5

Aim of Experiment: To design a web page using newly introduced HTML5 elements such as semantic tags, multimedia elements, form controls, and canvas.

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment – Web Browser, Text Editor, VSCode

Experiment Objectives:

- To understand semantic elements of HTML5.
- To implement audio, video, and canvas elements.
- To use new HTML5 form controls.
- To design a structured web page using HTML5.

Experiment Outcomes:

- Understanding of HTML5 Semantic Structure
- Implementation of Multimedia Elements
- Use of Graphics with Canvas
- Application of Advanced HTML5 Form Controls
- Development of a Complete HTML5 Web Page

Theory:

HTML5 introduced new elements that improve web structure, multimedia support, and user interaction such as semantic tags, multimedia elements, graphics, and advanced form controls.

Semantic Elements (Structure of Web Page)

1) Common Semantic Tags

- <header> – Defines the header section of a page or section.
- <nav> – Contains navigation links.
- <section> – Represents a thematic grouping of content.
- <article> – Represents independent, self-contained content.
- <aside> – Contains side content like ads or related links.
- <footer> – Defines the footer section.
- <main> – Represents the main content of the document.

2 Multimedia Elements (Audio & Video)

HTML5 provides built-in support for multimedia without plugins.

Tags

- <audio> – Embeds audio files.
- <video> – Embeds video files.
- <source> – Specifies multiple media sources.

3 Graphics Elements

HTML5 supports graphics and visual content.

Tags

- <canvas> – Used to draw graphics using JavaScript.
- <svg> – Used for scalable vector graphics.

4 Advanced Form Elements

HTML5 introduced new input types and form controls.

New Input Types

- email, url, number, date, time, range, color, tel, search

New Form Elements

- <datalist> – Provides predefined options.
- <progress> – Shows task progress.
- <meter> – Measures scalar values.
- <output> – Displays calculation results.

5 Interactive and Other Elements

Tags

- <details> – Creates expandable content.
- <summary> – Defines heading for <details>.
- <mark> – Highlights text.
- <figure> – Groups media content.
- <figcaption> – Caption for <figure>

Program Code:

```
<!DOCTYPE html>
<html>
<head>
<title>HTML5 New Elements Demo</title>
<style>
body { font-family: Arial; margin: 20px; }
header, footer { background: #2c7be5; color: white; padding: 10px; }
nav { background: #f1f1f1; padding: 10px; }
section { margin-top: 15px; padding: 10px; border: 1px solid #ccc; }
canvas { border: 2px solid black; }
</style>
</head>
<body>

<header>
<h1>HTML5 New Elements Example</h1>
</header>

<nav>
<a href="#">Home</a> |
<a href="#">About</a> |
<a href="#">Contact</a>
</nav>

<section>
<article>
<h2>1. Semantic Elements</h2>
<p>This page is designed using HTML5 semantic tags.</p>
</article>
</section>

<section>
<h2>2. Audio Element</h2>
<audio controls>
<source src="sample.mp3" type="audio/mp3">
Your browser does not support audio.
</audio>
</section>

<section>
<h2>3. Video Element</h2>
<video width="320" height="240" controls>
<source src="sample.mp4" type="video/mp4">
Your browser does not support video.
</video>
</section>

<section>
<h2>4. Canvas Element</h2>
```

```
<canvas id="myCanvas" width="200" height="100"></canvas>
<script>
var c = document.getElementById("myCanvas");
var ctx = c.getContext("2d");
ctx.fillStyle = "orange";
ctx.fillRect(20, 20, 150, 50);
</script>
</section>

<section>
<h2>5. HTML5 Form Controls</h2>
<form>
Name: <input type="text"><br><br>
Email: <input type="email"><br><br>
Date of Birth: <input type="date"><br><br>
Age: <input type="number" min="1" max="100"><br><br>
Satisfaction: <input type="range" min="1" max="10"><br><br>
<input type="submit" value="Submit">
</form>
</section>

<footer>
<p>HTML5 Experiment Demo</p>
</footer>
```

```
</body>
</html>
```

Task for Students: Design a “Student Information and Feedback Web Page” using newly introduced HTML5 elements.

Task Requirements:

- 1) Use HTML5 Semantic Elements
- 2) Use HTML5 Multimedia Elements
- 3) Use HTML5 Form Enhancements
- 4) Use HTML5 Graphics Element
- 5) Validation and Compatibility

Submission Requirement:

- 1) html5_experiment.html file
- 2) Screenshots of output in browser

Observations:

- Semantic HTML5 elements improved the structure and readability of the web page.
- Audio and video elements worked smoothly without external plugins.

- Canvas element enabled drawing and graphical content using JavaScript.
- New HTML5 form controls improved user input validation and interaction.

Conclusion:

This experiment successfully demonstrated the use of newly introduced HTML5 elements in designing a modern web page. By implementing semantic tags, the structure of the web page became more meaningful, organized, and easier to understand. The integration of multimedia elements such as audio and video enabled direct media playback in the browser without the need for external plugins, enhancing user experience. The use of the canvas element provided an understanding of how graphics and dynamic content can be created using HTML5 and JavaScript. Additionally, advanced HTML5 form controls improved data validation and user interaction, making the web page more efficient and user-friendly. Overall, the experiment enhanced practical knowledge of HTML5 and helped in understanding how modern web technologies contribute to better web design and functionality.

Expected Oral Questions:

1. Explain the HTML5 elements you used in your project?
2. How did you implement multimedia in your project?
3. How did you design the page layout using semantic tags?
4. What problems did you face while working with HTML5?
5. How does HTML5 improve user experience in your project?
6. How would you optimize an HTML5 web page?
7. How do you ensure browser compatibility in HTML5?

FAQs in Interview:

1. What is HTML5?
2. What are semantic elements?
3. Explain `<audio>` and `<video>`.
4. What is canvas?
5. What are new HTML5 input types?
6. Difference between `<div>` and `<section>`.
7. Role of HTML5 in modern web design.